

Bonus Homework 2

Discrete Mathematics by Hongfei Fu, 2023.10.23

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1. Consider a finite set X that contains n ($n \geq 1$) distinct elements. Prove that the maximum number of subsets contained in a collection $\mathcal{A} \subseteq 2^X$ such that (*) there are no subsets $A, B \in \mathcal{A}$ satisfying $A \neq B$ and $A \subseteq B$, is exactly $C_n^{\lfloor n/2 \rfloor}$ (or in a different notation, $\binom{n}{\lfloor n/2 \rfloor}$).

Hint: You may consider the following way. First prove the LYM-inequality which is stated as follows: for any finite set X with n elements ($n \geq 1$) and any collection $\mathcal{A} \subseteq 2^X$ satisfying the condition (*) above, we have that

$$\sum_{k=0}^n \frac{a_k}{C_n^k} \leq 1$$

where a_k is the number of subsets in $\mathcal{A}$ that has exactly k elements. The proof goes through a double-counting method that counts the number of permutations of n elements (i) through the permutation formula you have learned in your high school and (ii) through classifying each permutation into *disjoint* classes where each class contains all permutations whose prefix contains the same elements as some $A \in \mathcal{A}$. From the LYM-inequality, it is relatively easy to prove the Lemma.

Answer Area:

Lemma: the LYM-inequality

For any finite set X with n elements ($n \geq 1$) and any collection $\mathcal{A} \subseteq 2^X$ satisfying the condition (*) above, we have that

$$\sum_{k=0}^n \frac{a_k}{C_n^k} \leq 1$$

where a_k is the number of subsets in $\mathcal{A}$ that has exactly k elements.

proof:

Now we use two different ways to count the number of permutations of n elements.

- First, the number of permutations is $n!$.
- Next, suppose that $S \in \mathcal{A}$. S contains k elements.
 $\therefore S$ can be associated with some permutations whose prefix contains the same elements as S . The number of such permutations is $k!(n-k)!$.
 $\therefore \forall A, B, A, B \in \mathcal{A} (A \neq B), A \not\subseteq B$.
 $\therefore A, B$ are associated with different permutations.
 $\therefore$

$$\sum_{S \in \mathcal{A}} |S|!(n-|S|)! \leq n!$$

Suppose that a_k is the number of subsets in $\mathcal{A}$ that has exactly k elements. Then we have:

$$\sum_{S \in \mathcal{A}} |S|!(n-|S|)! = \sum_{k=0}^n a_k k!(n-k)! \leq n!$$

that is:

$$\sum_{k=0}^n a_k \frac{k!(n-k)!}{n!} = \sum_{k=0}^n \frac{a_k}{C_n^k} \leq 1$$

So the LMY inequality is proved. Now let's go back to the original problem.

$\therefore \forall k(0 \leq k \leq n), C_n^k \leq C_n^{\lfloor n/2 \rfloor}$.

$\therefore$

$$\sum_{k=0}^n \frac{a_k}{C_n^{\lfloor n/2 \rfloor}} \leq \sum_{k=0}^n \frac{a_k}{C_n^k} \leq 1$$

That is:

$$\sum_{k=0}^n a_k \leq C_n^{\lfloor n/2 \rfloor}$$

$\therefore$ The number of subsets contained in a collection $\mathcal{A}$ is $\sum_{k=0}^n a_k$

$\therefore$ the maximum number of subsets contained in a collection $\mathcal{A} \subseteq 2^X$ is exactly $C_n^{\lfloor n/2 \rfloor}$.